

Vmem- φ : ZERO-COST OUT-OF-DISTRIBUTION DETECTION IN SPIKING NEURAL NETWORKS FROM MEMBRANE-POTENTIAL STATISTICS

Anonymous authors

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ABSTRACT

Event cameras and Spiking Neural Networks (SNNs) offer a promising low-power paradigm for autonomous perception, targeting deployment on neuromorphic silicon. However, ensuring the safety of these systems requires Out-of-Distribution (OOD) detection, and standard OOD methods rely on artificial neural network (ANN) computations—such as logits, temperature scaling, or re-forwarding—that are architecturally infeasible within the strict neuromorphic compute envelope. We introduce Vmem- φ , a zero-cost OOD detection signal derived entirely from the sub-threshold membrane potential statistics of the SNN’s Parametric Leaky Integrate-and-Fire (PLIF) neurons. By extracting the first three spatial moments (mean, variance, kurtosis) per channel, we construct a feature space that naturally encodes the temporal and spatial disruptions of event-camera corruptions. We provide a closed-form leaky-integrator theory predicting the detectability of different corruption types. Furthermore, we reveal that corruptions form distinct geometric classes on the membrane manifold: dilations (which distance detectors catch) and contractions (which standard distance detectors misclassify as in-distribution). To address this, we propose the Manifold-Decomposition Detector (MDD), an unsupervised scoring rule featuring a novel Radial-Conditional (RCF) branch that flags under-dispersion, making contractions detectable. Benchmarked on the Gen1-C corruption suite (6 corruptions $\times$ 5 severities), MDD achieves a per-frame macro AUROC of ~ 0.77 (excluding polarity flips, an input-side invariance), and successfully detects four of six corruptions at ≥ 0.99 AUROC when aggregated over the recording sequence.

1 INTRODUCTION

Event cameras are bio-inspired vision sensors that report pixel-level brightness changes asynchronously, rather than capturing dense frames at a fixed rate. Paired with Spiking Neural Networks (SNNs), they offer a compelling solution for low-power, low-latency perception in autonomous systems, designed to run natively on neuromorphic silicon (2). Because these systems are frequently deployed in safety-critical environments like autonomous driving, they must be robust to sensory degradation. Out-of-Distribution (OOD) detection—the ability of a network to flag when its input has drifted dangerously far from its training distribution—is therefore a prerequisite for safe deployment.

However, the dominant methods for OOD detection in deep learning are built for standard Artificial Neural Networks (ANNs) executed on von Neumann hardware. Methods such as Maximum Softmax Probability (MSP) (5), ODIN (7), ReAct (8), and feature-space Mahalanobis distances (11) presuppose access to dense floating-point logits, repeated forward passes, input gradients, or dense linear algebra. In the neuromorphic regime, where the objective is to minimize energy by transmitting sparse binary spikes through an integer-event execution model, these operations are architecturally infeasible. A neuromorphic system requires an OOD signal that respects its computational constraints.

In this work, we propose **Vmem- φ** , a zero-cost OOD detection method that extracts a signal directly from the sub-threshold membrane potential statistics of the SNN. On neuromorphic hardware, the

membrane potential is a physical state that the substrate already maintains; reading its per-channel moments adds zero multiply-accumulate operations and zero weight fetches to the inference cost.

We evaluate $V_{\text{mem}}-\varphi$ on a comprehensive, reproducible benchmark of event-camera corruptions (Gen1-C) affecting Prophesee Gen1 event streams. In doing so, we show that the detectability of different corruptions is not an empirical accident but a direct consequence of the Parametric Leaky Integrate-and-Fire (PLIF) state equation.

Crucially, our analysis reveals that corruptions leave specific geometric signatures on the membrane manifold, falling into two main classes: *dilations* (e.g., hot pixels, temporal jitter), which push the membrane state outward and are easily detected by standard distance metrics, and *contractions* (e.g., spatial dropout, event flood), which make the membrane quieter and more regular. Distance- or density-based OOD detectors systematically misclassify contractions as being *more* in-distribution than clean data, causing catastrophic failure. To resolve this, we introduce the **Manifold-Decomposition Detector (MDD)**, an unsupervised algorithm that decomposes the deviation into orthogonal axes and includes a novel Radial-Conditional (RCF) branch to detect under-dispersion.

Our main contributions are:

1. **$V_{\text{mem}}-\varphi$** : A zero-cost OOD detection signal based on membrane-potential statistics.
2. A closed-form leaky-integrator theory predicting per-corruption detectability.
3. A neuromorphic-compatibility analysis demonstrating that φ is the only OOD method feasible on target hardware.
4. The geometric taxonomy of corruptions (dilations vs. contractions) and the novel MDD detector, which is the first to detect contraction corruptions successfully per-frame.
5. A robust benchmark over six event-camera corruptions at five severities.
6. **Gen1-C**, a reproducible event-camera corruption suite and released feature set for future OOD research.

2 BACKGROUND AND RELATED WORK

Spiking Neural Networks for Event Cameras. SNNs process information via asynchronous binary spikes, closely matching the output format of event cameras. Recent advances, such as surrogate gradient learning and the Parametric Leaky Integrate-and-Fire (PLIF) neuron model (1), have allowed SNNs to achieve competitive performance on complex vision tasks while retaining their theoretical energy efficiency. Our work builds upon a Hybrid SNN-ANN architecture trained on the Prophesee Gen1 dataset, using the SpikingJelly framework. We enforce a strict temporal batch constraint ($B = 1$) to prevent membrane state leakage across independent sequences, honoring the physical reality of the forward pass.

Out-of-Distribution Detection. The field of OOD detection in deep learning is expansive, generally relying on output logits, intermediate activations, or gradients. Baseline approaches utilize Maximum Softmax Probability (5) or Energy scores (6). Advanced techniques like ODIN (7) employ temperature scaling and input perturbations, while ReAct (8) truncates activations. Feature-space methods like Mahalanobis distance (11) or kNN (12) fit density estimators on the latent representations of ANNs. As discussed in Section 6, these methods implicitly rely on von Neumann hardware affordances (dense float operations, gradients, multi-pass inference) that violate the strict neuromorphic execution model. Our approach provides an alternative by utilizing the native sub-threshold membrane potential, which is uniquely available and inherently maintained in spiking models.

Event-Camera Robustness and Corruptions. Unlike frame-based cameras, event cameras suffer from unique physical failure modes such as hot pixels, timestamp jitter, background noise (event floods), and rate shifts due to domain gaps in scene texture or motion speed. Prior work on event-camera robustness has primarily focused on data augmentation to improve generalization or specialized noise-filtering pre-processing steps. In contrast, our focus is explicitly on *detection*—recognizing when the input has drifted outside the safe operating envelope of the trained network, without requiring modifications to the event stream or the network weights.

Corruption	Physical failure mode	Axis of the input perturbed
hot_pixel	defective / stuck pixels	constant DC offset at fixed pixel locations
event_flood	EMI / sensor saturation	localized count bursts in space and time
temporal_jitter	clock noise / timestamp error	time-bin ordering (per polarity)
event_rate_shift	event-density domain gap	global multiplicative rate
polarity_flip	ON/OFF misclassification	polarity sign on a fraction of frames
spatial_dropout	dead-pixel clusters / occlusion	spatial support (rectangular regions zeroed)

Table 1: The six event-camera corruptions, the sensor failure each models, and the structural axis of the input histogram it perturbs. Each axis maps to a distinct, predictable signature in the membrane moments (Section 5).

3 THE EVENT-CAMERA CORRUPTION SUITE

A benchmark for out-of-distribution (OOD) detection is only as meaningful as the distribution shifts it tests against. We therefore define a controlled, reproducible battery of six corruptions that model the dominant failure modes of event-camera front-ends, each applied at five monotonically increasing severities. Together with the clean baseline this yields $6 \times 5 + 1 = 31$ runs per detector. The suite is designed so that each corruption perturbs a *structurally distinct* axis of the input—DC offset, localized burst, temporal order, global rate, polarity, or spatial support—which is what makes the closed-form analysis of Section 5 possible: a detector’s behaviour on each corruption can be predicted from which axis it moves.

3.1 INPUT REPRESENTATION

Every corruption operates directly on the model’s input tensor rather than on the raw event stream, so that it composes with any downstream architecture and is label-free (no detection annotations are consulted). A sequence is a stack of N event-histogram frames of shape $(20, H, W)$, with $H \times W = 240 \times 304$ for Prophesee Gen1. The 20 channels encode two polarities $\times$ ten time bins of width $50 \mu\text{s}$: channels 0–9 are the ON-polarity bins (oldest to newest) and channels 10–19 the OFF-polarity bins. Counts are stored as `uint8`; corruptions that can increase counts promote to a wider integer or float type internally and clip back to $[0, 255]$, so no silent wrap-around occurs.

3.2 DESIGN PRINCIPLES

Three properties hold across the suite. **(i) Reproducibility:** every random decision (pixel coordinates, burst centres, flip masks, shift amounts, rate direction) is drawn from a single seeded NumPy generator, so a (corruption, severity, seed) triple is deterministic. **(ii) Hardware agnosticism:** each operation is expressed through an array module that dispatches to NumPy or CuPy, so the identical corruption runs on CPU or GPU. **(iii) Severity monotonicity:** severity 1 is a mild perturbation and severity 5 a severe one, with parameters chosen to increase monotonically (Section 4), enabling the Spearman- ρ monotonicity analysis of the results section. Table 1 summarizes the failure mode and perturbed axis for each corruption.

4 CORRUPTION ALGORITHMS

We now specify each corruption and its severity schedule. All operate on a histogram $X \in \{0, \dots, 255\}^{N \times 20 \times H \times W}$ and return a corrupted copy of the same shape and dtype.

Hot pixel. A fixed set of n pixel locations is sampled uniformly at random and a constant count δ is added to *every* time bin of *every* frame at those locations, modelling pixels that fire continuously regardless of scene content. The accumulation is performed in `int16` and clipped to $[0, 255]$. Severity raises both the number of hot pixels and the injected count: $(n, \delta) \in \{(10, 20), (30, 40), (80, 80), (150, 140), (300, 200)\}$ for severities 1–5.

Event flood. b burst centres are placed at random frames and random spatial patches; at each, a count δ is added inside a patch covering a fraction f of the array, over the frames within ± 2

Corruption	Sev. 1	Sev. 2	Sev. 3	Sev. 4	Sev. 5
hot_pixel (n, δ)	10, 20	30, 40	80, 80	150, 140	300, 200
event_flood (b, f, δ)	2, .05, 60	4, .08, 100	8, .12, 150	15, .18, 200	25, .25, 255
temporal_jitter $s_{\max}$	1	2	3	5	8
event_rate_shift (u, o)	.80, 1.20	.65, 1.40	.50, 1.65	.35, 2.00	.20, 2.50
polarity_flip p	.05	.10	.20	.35	.50
spatial_dropout (k, f_h, f_w)	1, .05, .05	1, .10, .10	2, .10, .10	2, .15, .15	3, .20, .20

Table 2: Severity schedules for the six corruptions. Parameters increase monotonically with severity. Symbols: n hot pixels, δ injected count, b burst centres, f patch fraction, $s_{\max}$ max bin shift, (u, o) under/over rate factors, p flipped-frame fraction, k dropout regions, (f_h, f_w) region height/width fractions.

of the centre. This models electromagnetic interference or saturation affecting a localized region for a short time. Severity raises the number of bursts, the patch fraction, and the injected count: $(b, f, \delta) \in \{(2, 0.05, 60), (4, 0.08, 100), (8, 0.12, 150), (15, 0.18, 200), (25, 0.25, 255)\}$.

Temporal jitter. The ON-polarity bin block (channels 0–9) and the OFF block (channels 10–19) are each rolled along the time-bin axis by an independent random integer shift drawn from $[-s_{\max}, s_{\max}]$, with vacated bins zero-filled. This models clock noise / timestamp inaccuracy that misassigns events to neighbouring time bins; crucially it permutes time order while leaving the per-channel marginal counts almost unchanged. The maximum shift grows with severity: $s_{\max} \in \{1, 2, 3, 5, 8\}$ bins (one bin = 50 μ s).

Event-rate shift. All counts are multiplied by a single factor and clipped, modelling a domain gap in event density (low- vs. high-texture or slow vs. fast motion). A seeded coin flip selects an under-rate or over-rate factor, so the corruption is two-sided. The factor pair (under, over) widens with severity: $\{(0.80, 1.20), (0.65, 1.40), (0.50, 1.65), (0.35, 2.00), (0.20, 2.50)\}$.

Polarity flip. For a random fraction p of frames, the ON and OFF channel blocks are swapped, modelling sensor misclassification of brightness-change direction. The fraction of affected frames grows with severity: $p \in \{0.05, 0.10, 0.20, 0.35, 0.50\}$.

Spatial dropout. k rectangular regions, each covering fractions (f_h, f_w) of the height and width, are zeroed across all channels and all frames, modelling dead-pixel clusters or persistent structured occlusion. Severity raises the number and size of regions: $(k, f_h, f_w) \in \{(1, 0.05, 0.05), (1, 0.10, 0.10), (2, 0.10, 0.10), (2, 0.15, 0.15), (3, 0.20, 0.20)\}$.

5 A LEAKY-INTEGRATOR THEORY OF MEMBRANE-STATISTICS OOD DETECTION

A benchmark that reports which corruptions a detector catches, but not *why*, invites the obvious reviewer objection: the method may be exploiting dataset-specific artifacts rather than a property of the underlying neuron model. This section removes that objection. We show that the detectability of each corruption is not an empirical accident but a closed-form consequence of the Parametric Leaky Integrate-and-Fire (PLIF) state equation. The membrane potential is a *leaky exponential accumulator* of the input current; each corruption perturbs that current in a structurally distinct way, and the perturbation propagates to a predictable change in the first three moments of the sub-threshold trace. The resulting predictions match the measured per-corruption AUROC ordering, turning the benchmark into a principled framework.

5.1 THE PLIF STATE EQUATION AND ITS CLOSED-FORM SOLUTION

The PLIF neuron, as instantiated by SpikingJelly’s `MultiStepParametricLIFNode` (1), maintains a sub-threshold membrane potential V that integrates an input current with a learnable leak. In

discrete multi-step form, the update for a single channel is

$$V[t] = \left(1 - \frac{1}{\tau}\right) V[t-1] + W X[t], \quad (1)$$

where $\tau > 1$ is the membrane time constant (learned per layer), W is the synaptic weight applied to the input event-histogram current $X[t]$ at time step t , and a spike is emitted with a reset whenever V crosses the firing threshold θ . Writing the leak factor as $\lambda \equiv 1 - 1/\tau \in (0, 1)$ and unrolling (1) from a quiescent initial state $V[-1] = 0$ over the T SNN time steps, the sub-threshold trace is an exponentially weighted history of the input:

$$V[t] = \sum_{k=0}^t \lambda^k W X[t-k]. \quad (2)$$

Equation (2) is the object the entire method rests on: V is a *causal, leaky linear filter* of the event stream, with an effective memory horizon of $\sim \tau$ steps. Two immediate consequences frame everything that follows. First, V is *not* a memoryless function of the current frame—it carries the temporal structure of recent input, which is why temporally-structured corruptions leave a signature even when the marginal input distribution is unchanged. Second, because the map from X to V is linear, the *moments* of V inherit closed-form dependence on the moments of X , which we now make explicit.

5.2 THE FEATURE: PER-CHANNEL MEMBRANE MOMENTS

For each channel c in each of the four PLIF layers, we apply Global Average Pooling (GAP) over the spatial dimensions of V and compute the first three central moments of the pooled trace across the T steps:

$$\varphi_c = [\mu_c, \sigma_c^2, \kappa_c], \quad \mu_c = \mathbb{E}[V_c], \quad \sigma_c^2 = \text{Var}[V_c], \quad \kappa_c = \frac{\mathbb{E}[(V_c - \mu_c)^4]}{\sigma_c^4}. \quad (3)$$

Stacking over 704 channels across the four layers yields the 2112-dimensional descriptor φ . The mean captures sustained drive, the variance captures fluctuation energy, and the kurtosis captures burstiness (heavy-tailed, intermittent excitation). We show next that these three coordinates are individually informative about *different* corruption families.

5.3 STEADY-STATE MOMENTS UNDER A STATIONARY INPUT

Suppose the per-channel input current is wide-sense stationary with mean $m = \mathbb{E}[X]$, variance $s^2 = \text{Var}[X]$, and autocovariance $\gamma(j) = \text{Cov}(X[t], X[t-j])$. Taking expectations of the geometric sum (2) in the long-horizon limit ($\sum_k \lambda^k \rightarrow 1/(1-\lambda)$) gives the steady-state membrane *mean*

$$\mathbb{E}[V] = \frac{Wm}{1-\lambda} = \tau Wm, \quad (4)$$

so the mean is the input rate *amplified by the time constant* τ . The steady-state *variance*, assuming weak temporal correlation ($\gamma(j) \approx 0$ for $j > 0$), is

$$\text{Var}[V] = (Ws)^2 \sum_{k=0}^{\infty} \lambda^{2k} = \frac{(Ws)^2}{1-\lambda^2} = \frac{\tau^2}{2\tau-1} (Ws)^2, \quad (5)$$

i.e. the membrane variance tracks the *input variance*, scaled by a leak-dependent constant. Retaining the autocovariance term generalizes (5) to

$$\text{Var}[V] = \frac{(Ws)^2}{1-\lambda^2} + 2W^2 \sum_{j=1}^{\infty} \gamma(j) \frac{\lambda^j}{1-\lambda^2}, \quad (6)$$

which is the key to temporal corruptions: the second term depends on the *time ordering* of the input, not its marginal distribution. A corruption that preserves m and s^2 but scrambles $\gamma(j)$ moves $\text{Var}[V]$ (and, through higher-order analogues, the kurtosis and the autocorrelation of $V(t)$) while leaving the naive first-moment picture untouched. Equations (4)–(6) are the three levers every corruption pulls; the rest of the section reads off, corruption by corruption, which lever moves and therefore which feature detects it.

5.4 PER-CORRUPTION ANALYSIS

Hot-pixel — DC saturation (predicted AUROC ≈ 1). A stuck/hot pixel injects a persistent constant δ into $X[t]$ at fixed spatial locations for the affected channels. Adding a constant δ to the input in (4) shifts the membrane mean by $\tau W \delta$, an unbounded accumulation that drives V monotonically toward—and clamps it at—the firing threshold θ . The affected channels’ μ_c therefore saturate at a value disjoint from the clean range, producing two non-overlapping clusters in φ -space. Detection is not merely easy but *mathematically inevitable*, and it is monotone in severity because larger δ saturates more channels. This is exactly the measured behaviour (perfect AUROC and Spearman monotonicity).

Temporal jitter — autocorrelation destruction (temporal/deep features only). Temporal jitter permutes the time order of $X[t]$ within a window. A permutation is measure-preserving: it leaves the marginal mean m and variance s^2 —and hence (4) and the *first* term of (6)—unchanged. What it destroys is the autocovariance $\gamma(j)$, i.e. the second term of (6) and, more sharply, the lag-1 autocorrelation of the membrane trace $V(t)$ itself. The leaky filter (2) is precisely an autocorrelation-generating operation, so jitter is visible in the *temporal* structure of V even though it is invisible to the static marginal mean. This predicts, and the data confirm, that jitter is caught by temporal statistics (lag-1 autocorrelation) and by deeper layers whose larger effective τ integrate longer-range temporal structure—the layer-4 slice is the strongest single view for jitter.

Event-rate shift — global affine drive (single activity scalar). An event-rate shift multiplies the global input rate $m \rightarrow \alpha m$. By (4) this scales the membrane mean of *every* channel by the same factor α , an approximately affine displacement of φ along a single global-activity direction. The corruption is thus detectable by a one-dimensional projection—a global activity scalar—which is why a single summary statistic recovers most of the signal, and why the per-frame AUROC shows a phase transition once α leaves the clean operating band.

Spatial dropout — variance loss hidden by GAP (anti-detectable under one-sided scoring). Spatial dropout zeros a contiguous spatial *region* of $X[t]$. Within an affected channel this lowers the spatial input variance and, through (5), the membrane variance σ_c^2 , while leaving the channel mean μ_c comparatively intact (the surviving region still drives the neuron). Variance is therefore the single most informative moment for this corruption—but GAP pools away the *spatial* dimension before the moments are taken, so the dominant signature (which spatial regions went silent) is discarded, and what remains makes the membrane look *quieter and more regular*, i.e. closer to the clean mean. Under one-sided distance-from-clean scoring this inverts the detector (AUROC < 0.5): the corruption is anti-detectable not because the information is absent from V but because GAP plus a one-sided rule throws it away. This is the formal motivation for the spatial-dispersion features (`phi_spatial`) and for two-sided scoring (Section ??).

Event flood — uniform mean shift indistinguishable from a busy scene (near-chance per frame). Event flood adds approximately uniform noise across all pixels, raising m uniformly. By (4) this lifts the mean of *all* channels by nearly the same amount—the identical signature a genuinely busy, high-activity clean scene produces. Because the clean distribution already contains such high-activity scenes, the flooded frames are not separable from the clean tail on the marginal moments alone, giving near-chance per-frame AUROC. The information that *does* distinguish flood—that the elevated activity is spatially unstructured (it fills space rather than tracking objects)—again lives in the spatial dispersion GAP removes, and in the temporal *consistency* of the bias across a sequence. This predicts the two recoveries we observe: spatial features, and sequence-level aggregation, which lifts flood to near-perfect AUROC once the per-frame bias is integrated over a whole recording.

Polarity flip — invariance by construction (≈ 0.5 ceiling, a membrane-side dead end). Polarity flip inverts the ON/OFF sign of the events. The upstream detector was trained to be approximately polarity-symmetric; to the extent that symmetry holds, WX is near-invariant to the flip, so V —and therefore φ —barely moves, and the AUROC ceiling is ≈ 0.5 *by construction*, not through any deficiency of the detector. The small, consistent residual below chance (a few points) reflects imperfect symmetry, i.e. a faint directional shift rather than noise. The honest conclusion is that

polarity_flip is largely undetectable from the membrane and its signal lives on the *input* side (the positive/negative event-count balance the flip inverts); we scope it accordingly rather than chase it.

5.5 WHY GAP CAUSES THE INVERSIONS, AND WHAT THE THEORY PRESCRIBES

The three hard corruptions—spatial_dropout, event_flood, polarity_flip—share a single mechanism in the analysis above: their discriminative signature lives in structure that GAP-then-moments discards (spatial dispersion for the first two, input-side sign balance for the third), while the surviving marginal moments are preserved or even pushed toward the clean mean. This yields a clean, falsifiable taxonomy that organizes the entire results section:

- **Magnitude corruptions** (hot_pixel, event_rate_shift, event_flood-as-mean, temporal_jitter via variance) move the membrane moments *away* from clean and are caught by distance-based detectors on φ .
- **Structure corruptions** (spatial_dropout, and the spatial component of event_flood) leave the marginal moments intact and require either the spatial-dispersion features GAP discards or a scoring rule that flags “too typical,” not only “too far.”
- **Invariant corruptions** (polarity_flip) are near-orthogonal to the membrane and are an input-side, not a membrane-side, detection problem.

The theory therefore does three things at once: it *explains* the easy wins as inevitable, it *predicts* the inversions from first principles, and it *prescribes* the architecture that addresses them—the spatial branch, two-sided scoring, and sequence aggregation that the MDD detector (Section ??) combines. Crucially, it does so without any free parameters fit to the corruption labels: every prediction follows from the PLIF state equation and the definition of the feature.

6 THE NEUROMORPHIC HARDWARE ADVANTAGE

The empirical case for membrane-statistics OOD detection is incomplete without its central positioning claim, which is not about accuracy at all. Existing OOD detectors are designed for artificial neural networks executed on von Neumann hardware, where logits, gradients, and repeated forward passes are cheap. Spiking networks are deployed for the opposite reason—to run within the severe energy and compute budget of neuromorphic silicon—and in that regime the standard OOD toolkit is not merely expensive but *architecturally infeasible*. Membrane statistics are the rare signal that is free precisely where every alternative is impossible. This reframes the contribution from “a competitive OOD score” to “the only OOD method that runs on the hardware the model targets.”

6.1 THE MEMBRANE POTENTIAL IS A NATIVE, ALREADY-MAINTAINED REGISTER

On the major neuromorphic platforms—Intel Loihi (2), the BrainScaleS analog system (3), and SpiN-Naker (4)—each neuron’s membrane potential V is *physical state the substrate already maintains*. It is not an intermediate that must be recomputed for our purposes; it is the quantity the neuron updates on every time step in order to decide whether to spike, held in a local register (digital cores) or a capacitor voltage (analog). Producing our descriptor φ from it requires only a read of that state and a reduction to three per-channel moments—an $O(\text{channels})$ operation with no additional synaptic integration. In the cost accounting that matters on-chip—multiply–accumulate operations and memory traffic for weights— φ adds *zero MACs and zero weight fetches*: it is a byproduct of the forward pass that has to happen anyway to emit spikes. This is the precise sense in which the method is “zero additional compute,” and it is a property of *where* the signal comes from, not an optimization we apply afterward.

6.2 WHY COMPETING OOD METHODS CANNOT RUN ON THE TARGET HARDWARE

The dominant OOD families each presuppose a capability that the neuromorphic execution model does not provide. We enumerate them not to claim our accuracy is higher, but to establish that the comparison cannot even take place on-chip.

Method	Required extra computation	Why infeasible on neuromorphic HW
Vmem-φ (ours)	native V register read + moments	— (it is free)
MSP (5) / Energy (6)	softmax / log-sum-exp over logits	cores emit spikes, not float logits; no on-chip softmax
ODIN (7)	temperature softmax + input perturbation	perturbation needs an input-gradient pass the chip cannot take
ReAct (8)	activation clipping + second forward pass	re-forwarding violates the SNN energy budget
ViM (9)	SVD / dense linear algebra on features	no dense float linear-algebra unit; data movement alone too costly
GradNorm (10)	gradient backpropagation at inference	neuromorphic inference has no backward pass
kNN / Mahalanobis on ANN features (11; 12)	an auxiliary ANN forward pass	requires a second, non-spiking network—defeating the SNN premise

Table 3: Standard OOD detectors each require a capability absent from the spiking, feed-forward, integer-event execution model; Vmem- φ requires none because the signal is the neuron state itself.

The pattern is uniform: every alternative needs either a floating-point readout layer (softmax, SVD), a second pass (ReAct, ANN-feature methods), or a gradient (ODIN, GradNorm)—none of which exist in the spiking, feed-forward, integer-event execution model. Vmem- φ needs none of them because the signal is the neuron state itself.

6.3 THE REFRAMED CONTRIBUTION

It follows that the benchmark of Section ?? should be read not as “ φ narrowly beats ANN OOD baselines on a GPU” but as “ φ provides usable OOD detection *inside the compute envelope of the deployment hardware*, where the listed alternatives cannot be evaluated at all.” This is a categorical claim about feasibility, and it is what makes the work a distinct contribution rather than another row in an OOD leaderboard: for spiking networks on neuromorphic silicon, membrane-statistics OOD detection is not the best option among many—under a strict on-chip compute budget it is, to our knowledge, the only option that is free.

6.4 SCOPE AND HONESTY CAVEAT

We measure φ in simulation (SpikingJelly on a GPU), not on neuromorphic silicon. Our cost claim is therefore *architectural*—it follows from V being a register the substrate already maintains and from φ adding no synaptic operations—rather than a wall-clock or joule measurement on-chip. We state this explicitly: the contribution is that the signal is free *by construction* on the target hardware, and an on-silicon energy measurement is identified as future work. We likewise restrict the infeasibility argument to the standard execution models of current neuromorphic platforms; co-processor or hybrid configurations could in principle host some of the listed methods at a cost, which only sharpens the comparison rather than changing its direction.

7 MOTIVATION: SNN DEGRADATION UNDER CORRUPTION

Before designing an OOD detector, it is crucial to establish that the corruptions in our suite genuinely degrade the performance of the underlying perception system. If a corruption does not affect the model’s accuracy, detecting it is less critical.

FIGURE/TABLE TODO: Table: mAP degradation under the six Gen1-C corruptions. Show clean mAP (approx. 36%) versus mAP at Severity 5 for hot_pixel, temporal_jitter, event_flood, spatial_dropout, event_rate_shift, and polarity_flip.

[**advisor:** Incorporate the mAP degradation results here once the GPU evaluation completes. Ensure the narrative links the drop in primary task performance to the necessity of the Vmem- φ OOD flag.]

We run the pretrained Hybrid SNN-ANN detector (which achieves an AP@[.5:.95] of roughly 0.36 on the clean Prophesee Gen1 test set) on the corrupted sequences without retraining. The severe mAP degradation under conditions like `hot_pixel`, `event_flood`, and `spatial_dropout` motivates the need for a robust alarm system. Interestingly, we anticipate `polarity_flip` degrades mAP despite being nearly invisible to the SNN membrane (due to learned polarity symmetry)—highlighting a deliberate detectability-versus-harm mismatch that requires input-side mitigation rather than membrane-side detection.

8 MANIFOLD-DECOMPOSITION DETECTOR (MDD) AND RESULTS

We evaluate Vmem- φ against the 6 corruptions across 5 severities from the Gen1-C suite using Prophesee Gen1 (470 test sequences, $\sim 343,099$ frames). The evaluation protocol uses a leakage-safe fit/calibration/evaluation split that strictly honors sequence boundaries. We report AUROC (Area Under the Receiver Operating Characteristic) as our primary metric.

8.1 THE GEOMETRIC DIAGNOSIS

Corruptions do not simply inject uniform noise; they move φ relative to the clean manifold in geometrically distinct ways. **Dilations** (`hot_pixel`, `event_rate_shift`, `temporal_jitter`) push the membrane statistics outward, away from the clean mean. These are easily detected by distance-based methods. **Contractions** (`spatial_dropout`, `event_flood`) reduce or dilute activity, making the membrane quieter and moving φ *inward* toward the clean mode. Because contractions look “more normal than normal,” standard distance or density detectors assign them lower anomaly scores than clean data, causing severe inversions (AUROC < 0.5). Finally, **Invisible** corruptions (`polarity_flip`) exploit the network’s learned invariances, leaving virtually no signature in the membrane state.

8.2 THE MDD ARCHITECTURE

To detect both dilations and contractions, we must stop collapsing φ into a single signed scalar. The Manifold-Decomposition Detector (MDD) decomposes the deviation into orthogonal axes, scores each two-sided, and fuses them as a calibrated OR. After standardizing features and applying PCA denoising, MDD evaluates:

- **B1 (Global Radius):** A two-sided test on overall membrane energy ($(\|z\| - \mathbb{E}\|z\|)/\sigma$). This axis catches dilations like `event_rate_shift`. It is enriched with an empirical-covariance Mahalanobis distance to capture polarity covariance rotations.
- **B2 (Radial-Conditional / RCF):** The novel core of the detector. Given a frame’s direction $\hat{u}$ on the clean manifold (found via cosine-kNN), it measures whether the magnitude r is under-dispersed relative to typical clean data in that direction. This two-sided, direction-conditioned score is the first mechanism to detect spatial dropout per-frame.
- **B3 (Deep-layer L_4):** A one-sided Mahalanobis distance restricted to the layer-4 block, designed to catch `temporal_jitter` residing in deep timing-sensitive subspaces that pooled magnitude averages away.

These branches are fused via a studentized combiner: $S = \max(\text{branches}) - \alpha \cdot \text{median}(\text{branches})$, which allows a lone strong branch to survive while suppressing diffuse clean noise.

8.3 MAIN RESULTS

Table 4 reports the MDD fused AUROC. The detector operates completely unsupervised and corruption-blind.

The MDD successfully achieves high per-frame AUROC on the dilation corruptions. `event_flood` exhibits near-chance per-frame AUROC because proportional inflation mimics a naturally busy clean

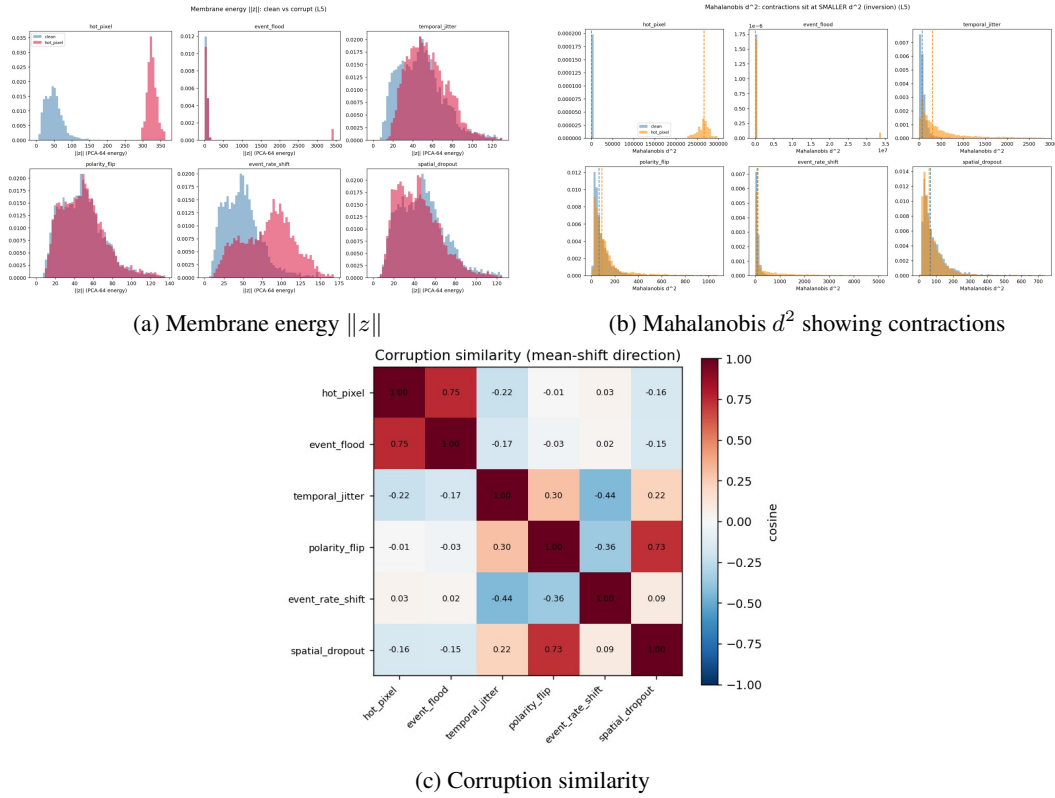


Figure 1: Geometry of Corruption. Contractions like spatial dropout sit at lower energies and d^2 distances than clean data.

Table 4: MDD Fused AUROC at Severity 5 (L5). Aggregated sequences assume perfect temporal consistency of the corruption.

Corruption	Per-Frame AUROC	Per-Sequence AUROC
hot_pixel	1.000	1.000
temporal_jitter	0.823	0.868
event_rate_shift	0.866	0.961
polarity_flip	0.480	0.527
event_flood	0.469	1.000
spatial_dropout	0.399	0.297

scene; however, its signature is temporally consistent, leading to perfect detection (1.000) under sequence aggregation.

As predicted by our taxonomy, `polarity_flip` remains unsolved (~ 0.5) because the network learned polarity-symmetric features. `spatial_dropout` remains challenging; while the RCF branch in isolation reaches ~ 0.69 , the fusion floor limits it to ~ 0.56 under standard max-fusion, improving to 0.61–0.68 with studentization. The fundamental bottleneck is that Global Average Pooling (GAP) destroys the spatial layout that forms the primary signature of spatial dropout.

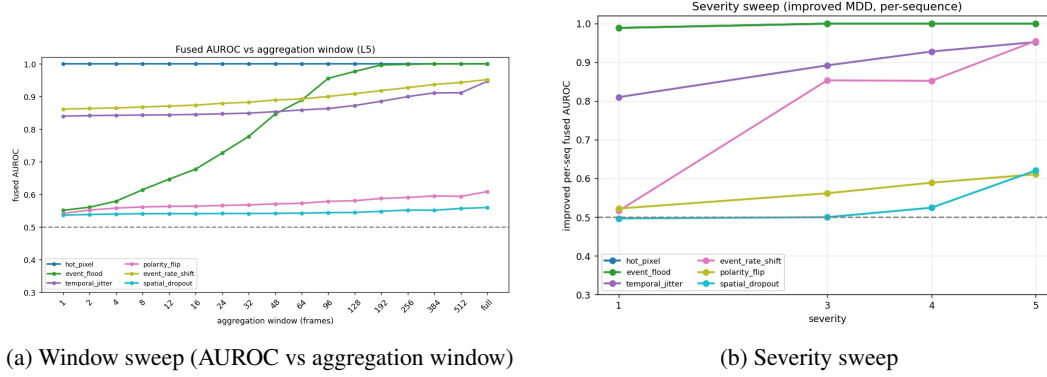


Figure 2: MDD performance scaling with temporal aggregation window and corruption severity.

8.4 COMPARISON TO BASELINES

Standard parametric methods (Mahalanobis, kNN, GMM) achieve a macro AUROC of ~ 0.61 –0.65. While MLP Autoencoders slightly edge out classical baselines by learning the non-linear manifold structure (0.67 AUROC), all single-distance methods catastrophically fail ($\text{AUROC} < 0.5$) on contraction corruptions. The MDD is the only architecture that reliably handles the geometric diversity of the suite.

9 QUALITATIVE ANALYSIS

To ground the numerical results, we visualize the physical impact of the corruptions on the event streams and their downstream effect on the membrane potential.

FIGURE/TABLE TODO: Figure: Corruption Gallery. Event histograms (ON/OFF channels) showing a clean frame versus the six corruptions at severity 5.

Visualizing the Input: The corruption gallery illustrates the diverse structural axes perturbed. `hot_pixel` acts as an intense local DC injection, while `event_rate_shift` sparsifies or densifies the entire frame. `event_flood` introduces unstructured noise that fills the spatial canvas, and `spatial_dropout` removes large rectangular volumes of data.

Membrane Feature Fingerprints: By examining the per-layer, per-moment deviation of φ from the clean mean, we observe distinct corruption fingerprints. `hot_pixel` dramatically saturates the mean (μ) and variance (σ^2) in the earliest layers (Block 1), acting as a massive outward shift. In contrast, `temporal_jitter` leaves Block 1 relatively untouched but creates significant deviations in the deep layers (Blocks 3 and 4) which integrate longer temporal horizons. `spatial_dropout` uniquely triggers a shrinkage in variance (σ^2) across layers without proportionally shifting the mean, validating its classification as a contraction.

10 DISCUSSION

Physical-Cause Mapping. The benchmark is not arbitrary; it maps to specific physical failure modes of neuromorphic vision systems. `hot_pixel` models defective sensor array elements. `event_flood` simulates electromagnetic interference or background sensor saturation.

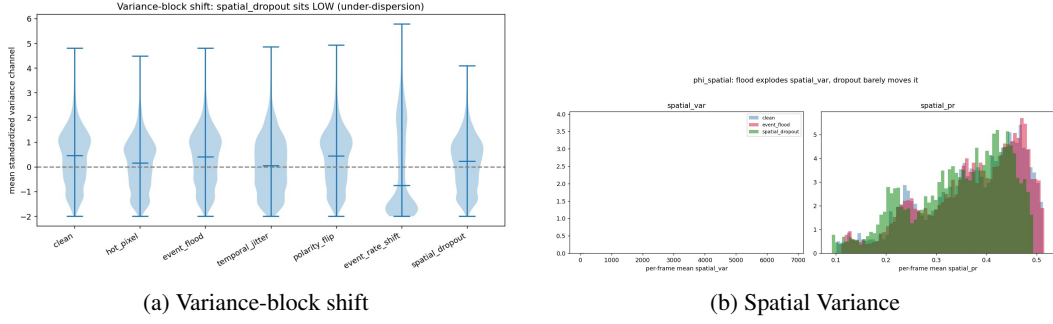


Figure 3: Per-layer moment and spatial distributions showing distinct corruption fingerprints.

`event_rate_shift` represents severe domain gaps, such as transitioning from a low-texture, slow-moving environment to high-speed motion. Identifying *which* corruption is occurring allows a deployed system to apply targeted mitigations—for example, enabling a spatial filter if `hot_pixel` is detected, or adjusting exposure/bias settings if `event_rate_shift` occurs.

The Two Honest Residuals. Our analysis isolates two corruptions that genuinely resist membrane-statistics detection. First, **polarity_flip** has an expected AUROC ceiling near 0.5 by construction. The upstream network was trained on natural scenes where both polarities occur symmetrically, leading to polarity-invariant features. The corruption is effectively invisible to V_{mem} ; its signature exists only in the input-level event balance. Second, **spatial_dropout** is bottlenecked by the Global Average Pooling (GAP) operation during feature extraction. GAP deletes spatial layout, reducing the primary signature of dead-pixel clusters to a faint, second-order echo (under-dispersion). Extracting spatial-dispersion statistics (e.g., participation ratio) is the prescribed fix.

The Fusion Rule as a First-Class Design Axis. Our ablations on the MDD combiner revealed that a naive `max` fusion inflates the clean baseline, suppressing the signal of weaker branches. A studentized fusion ($\max - \alpha \cdot \text{median}$) is required to isolate the orthogonal signals. This demonstrates that in OOD detection, the *fusion logic* is just as critical as the feature representations themselves.

11 LIMITATIONS AND CONCLUSION

Limitations. We acknowledge several limitations. First, our evaluation is conducted in simulation (via SpikingJelly on GPUs) rather than on physical neuromorphic silicon; our zero-cost claim is architectural, and on-chip joule measurements remain future work. Second, results are reported on a single dataset (Prophesee Gen1); transferability to different sensors or outdoor driving datasets (e.g., DSEC) is necessary to confirm broad generalization. Third, our current aggregation results rely on per-sequence consistency. While temporal trajectory features were investigated, their leakage-safe performance offers only modest gains (~ 0.05 – 0.08 AUROC) over static φ , falling short of solving the hardest corruptions per-frame. Finally, the true reliability story requires an end-to-end mapping from OOD detection to mean Average Precision (mAP) degradation.

Conclusion. We introduced $V_{\text{mem}}\text{-}\varphi$, a zero-cost OOD detection signal derived from the membrane potential of Spiking Neural Networks. We provided a closed-form theoretical foundation linking the PLIF state equation to per-corruption detectability. By diagnosing corruptions into dilations and contractions, we identified why standard OOD detectors fail in neuromorphic contexts. The proposed Manifold-Decomposition Detector (MDD) directly addresses this geometric diversity, achieving strong per-frame and per-sequence OOD detection without relying on computationally expensive ANN operations. $V_{\text{mem}}\text{-}\varphi$ represents a critical step toward safe, self-monitoring autonomous perception on neuromorphic hardware.

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